What Does This Test Support?

AP Statistics · Unit 4 · Topic 4.10 · B: 2027-01-29 · E: 2027-03-19 · TEACHER KEY

CED TETHER

- **Skill 3.E** — Calculate a test statistic and find a p-value, provided conditions for inference are met.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.E** — Justify a claim using a decision based on significance tests.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.I** — Calculate an appropriate test statistic for a difference of two means. [Skill 3.E]
- **EK VAR-7.I.1** — For a single quantitative variable, data collected using independent random samples or a randomized experiment from two populations, each of which can be modeled with a normal distribution, the sampling distribution of t = $[(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)] / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$ is an approximate t-distribution with degrees of freedom that can be found using technology. The degrees of freedom fall between the smaller of n_1 - 1 and n_2 - 1 and $n_1 + n_2$ - 2.
- **EU DAT-3** — Significance testing allows us to make decisions about hypotheses within a particular context.
- **LO DAT-3.G** — Interpret the p-value of a significance test for a difference of population means. [Skill 4.B]
- **EK DAT-3.G.1** — An interpretation of the p-value of a significance test for a two-sample difference of population means should recognize that the p-value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population means are equal to each other.
- **LO DAT-3.H** — Justify a claim about the population based on the results of a significance test for a difference of two population means in context. [Skill 4.E]
- **EK DAT-3.H.1** — A formal decision explicitly compares the p-value to the significance α. If the p-value $\leq \alpha$, then reject the null hypothesis, H_0: $\mu_1 - \mu_2 = 0$, or H_0: $\mu_1 = \mu_2$. If the p-value $> \alpha$, then fail to reject the null hypothesis.
- **EK DAT-3.H.2** — The results of a significance test for a two-sample test for a difference between two population means can serve as the statistical reasoning to support the answer to a research question about the populations that were sampled.

Essential question: How can one test support a mean difference without settling size, individuals, or cause?

DOK-3 skill: 4.E · **FRQ pattern:** `significance-vs-size-cause-and-scope`

DOK rationale: Part (c) synthesizes a warranted report from two incomplete accounts by coordinating the p-value decision, practical benchmark, mean-versus-individual scope, sampling, and self-selection.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Ask students to mark defensible and unsupported clauses in both reports.
- After the video, require numerical, design, and scope evidence.
- Collect and score part (c) only, E/P/I.

Students do

- Commit to a clause-level comparison, then complete the 7.9 follow-along.
- Synthesize one warranted report in (c), revise, and turn in.

Questions to ask

- Assuming what is 0.0263 calculated?
- Which benchmark did the test use: 0 or 3?
- What do sampling and self-selection each permit?

Adult role
Circulate. Ask students to preserve valid clauses from both reports.

[WARN] WATCH FOR

- Reading the p-value as $P(H_0)$.
- Equating significance with a 3-minute effect.
- Inferring cause from self-selected modes.

SCORING PART (c) — E / P / I

Expected elements

- **uses-test-decision** (required) — Retains Nia's p-value decision and positive mean conclusion
- **bounds-size-and-individuals** (required) — Uses 1.2 versus 3 and distinguishes means from individuals
- **bounds-scope-and-cause** (required) — Uses SRS scope and self-selection to reject causation
- **synthesizes-and-traces-risk** (required) — Corrects both reports and states a consequence of keeping either whole

E	Synthesizes the valid clauses, uses both numerical benchmarks, bounds population and causal scope, and states both decision risks.
P	Reaches the bounded conclusion but incompletely treats size, individuals, scope, cause, or one report's risk.
I	Retains H_0, treats significance as a 3-minute or individual guarantee, or infers cause from self-selection.

Common mistakes

- Reading 0.0263 as $P(H_0)$
- Treating a test against 0 as a test against 3
- Inferring cause from mode choice

[STAR] TODAY'S PROBLEM — annotated

Suppose 4,000 active students chose Focus mode and 3,600 chose Guided mode on an online practice service this term. Independent SRSs of 100 nonoverlapping students completed the same set; summaries are shown. Before sampling, the service asked whether $\mu_F > \mu_G$ and set $\alpha = 0.05$. Students chose their modes, and only a difference of at least 3 minutes is practically important.

Nia: "Since $p \approx 0.0263 < 0.05$, there is evidence that the Focus population mean is higher. Therefore Guided mode causes every student to finish at least 3 minutes faster; switch everyone."

Mateo: "The observed gap is only 1.2 minutes and students selected their modes, so neither a 3-minute effect nor causation is established. Therefore retain H_0; there is no evidence of a population mean difference."

Practice time summaries (minutes)

Mode	N	n	$\bar{x}$	s
Focus	4,000	100	42.8	4.2
Guided	3,600	100	41.6	4.5

FIRST TAKE (5 min)

Which clauses in Nia's and Mateo's reports look defensible? Cite one number and one design feature.
Key: Not graded. Credit a supported comparison of at least two clauses.

Rules callout "CARRY OUT THE TEST, THEN BOUND THE CONCLUSION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define μ_F and μ_G, then state H_0 and H_a in Focus-minus-Guided order.

Key: μ_F and μ_G are the mean completion times for all active Focus and Guided choosers this term. $H_0 : \mu_F - \mu_G = 0$ and $H_a : \mu_F - \mu_G > 0$.

DOK 2 **(b)** Check the two-sample t conditions. Compute the difference, SE, t, technology degrees of freedom, and one-sided p-value; interpret the p-value.

Key: The groups are independent SRSs; $100 \le 0.10(4{,}000) = 400$, $100 \le 0.10(3{,}600) = 360$, and both sample sizes exceed 30. The difference is 1.2 minutes. $SE = \sqrt{4.2^2/100 + 4.5^2/100} = 0.6155485$, $t = 1.9494807$, Welch $df \approx 197.065$, and the upper-tail $p \approx 0.0263285$. If H_0 is true, the chance of a statistic at least 1.9495 in the stated direction is about 0.0263.

DOK 3 **(c)** Evaluate both reports and write one warranted conclusion. Coordinate p versus α, the 1.2-minute gap versus the 3-minute benchmark, means versus individuals, SRS population scope, and self-selection. Explain the decision risk of retaining either report unchanged.

Key: Nia correctly uses $0.0263 < 0.05$: reject H_0 and report evidence that $\mu_F > \mu_G$ for this term's choosers. Mateo correctly rejects the practical and causal claims, but his retain-H_0 conclusion ignores the test result. The test compares the mean difference with 0, not 3, and the observed gap is only 1.2 minutes; it establishes neither a 3-minute effect nor that every Guided student is faster. The SRSs support inference to this term's two chooser populations, but self-selection is not random assignment, so causation is unsupported. Keeping Nia's report could prompt an unsupported universal switch; keeping Mateo's could discard real evidence of a mean difference.

EXIT

One thing the video changed about my first take: ______________________________